

Alpilla 600 MW Combined Cycle Power Plant

Document numbering and document control procedure

Document number	ALP-EPC-00000-GE-PRC-0002
Revision	draft - not yet issued (next issue: IFR rev A)
Originator	Istanbul EPC
Review class	approval
Planned issue (workflow timeline)	IFR 2027-04-07, IFA 2027-05-05, IFC 2027-05-12
Basis	requirement:ER-17.01

1. Purpose, scope and basis

This procedure sets the numbering of all project documents (consortium, Imaginary Electric, subsuppliers), the document types and their review class, revisions and purposes, the review cycle with the Owner and the maintenance of the master document list (MDL).

ER-17.01: Documents submitted electronically with the project numbering (consortium proposal adopted by the Owner in TQ-035: ALP-<ORG>-<KKS>-<DISC>-<TYPE>-<NNNN>) and a master document register; review period 14 days (21 days for the Basic Design Package).

- TQ-035 (closed, answered 2026-09-28 by Owner): (1) Accepted: the consortium numbering is the project numbering under ER-17.01; the Owner's EDMS will use it unchanged. (2) The KKS key list is approved for engineering. The Owner's Engineer verifies each key title against the VGB-S-811 key catalogue with the KKS manual review; until then keys stay flagged 'not verified'. (3) MDL accepted as baseline for the Basic Design Package. Review classes accepted with one change: logic diagrams (LOG) incl. cause and effect are safety relevant and are for approval. The MDL is updated monthly with the progress report. (4) AWP definitions accepted. The Owner requires every CWP to be released for construction only when its EWP is complete (all IFC) and its PWP materials are on site or have a confirmed delivery before the planned installation.
- TQ-IEC-023 (closed, answered 2026-09-28 by Imaginary Electric): (1) Confirmed. IEC tags the power island with the project KKS to component level (B), systems and equipment units per the project key list; IEC part numbers are kept as vendor attributes below component level. IEC checks the key titles of its main groups against its VGB-S-811 licence copy and reports deviations with the KKS manual review (IFR). (2) Confirmed: project number primary, IEC number as vendor reference on the title block. (3) Confirmed: the VDRL dates in the MDL are the IEC commitment. (4) Confirmed with an improvement: generator shop tests are brought forward; the GT generators leave the works in week 49 together with the GT (sea transport 4 weeks), on site 2027-11-08. The ST generator follows the ST train (no change).

2. Document number

Format: ALP-<ORG>-<KKS>-<DISC>-<TYPE>-<NNNN>

Field	Format	Meaning
ALP	fixed	Project code
ORG	3 characters	Originator (section 3)
KKS	NNAAA / 00000	KKS system G F0 F1F2F3 of the main system the document belongs to (ALP-EPC-00000-GE-PRC-0001); 00000 for plant-general documents
DISC	2 letters	Discipline (section 3)
TYPE	3 characters	Document type (section 4); defines the IEC 61355 class and the workflow rules
NNNN	4 digits	Running number per ORG-KKS-DISC-TYPE, from 0001; never reused

Example ALP-C03-00UPA-GE-ITP-0001: originator C03 (subsupplier); KKS 00UPA (CW intake and outfall structures (marine)); discipline GE (general); type ITP; number 0001

Sheets of a multi-sheet drawing share the number (sheet count in the MDL); a supplier's own number is recorded as vendor reference and shown as a secondary number in the title block (TQ-IEC-023).

3. Originator and discipline codes

ORG	Originator
EPC	Istanbul EPC (consortium leader)
IEC	Imaginary Electric (consortium member)

ORG	Originator
OWN	Owner
xxx	Subsupplier: 3-character code allocated by Istanbul EPC with the purchase order

DISC	Discipline
GE	general
PR	process
ME	mechanical
PI	pipng
EL	electrical
IC	instrumentation and control
CV	civil
SS	structural
HV	hvac
HS	hse
CM	commissioning
PC	project control

4. Document types and workflow rules

Prep = working days from inputs ready to first issue (longer for multi-sheet documents, at most double); Update = working days to incorporate comments; Inputs at = maturity the inputs must have reached (IFR or IFC); C = construction document (belongs to an EWP), P = procurement document (belongs to an MR).

Share = AWP progressive release: a construction document of this type is needed at CWP start - lead + share x CWP duration (0 = at the CWP start).

Type	Title	IEC 61355	Review class	Prep	Upd.	Inputs at	Typical inputs	C/P	Share
ARC	Architectural drawing	L	approval	20	10	IFR	LAY	C	0.3
BLK	Block / architecture diagram	F	approval	15	10	IFR	DBR, IOL, SLD		
CAL	Calculation	E	review	15	5	IFR	CAL, DBR, HMB, LST, PID, RPT, SDS, SLD, STU		
CBL	Cable schedule and routing	M	information	15	5	IFR	SLD, LAY	C	0.3
CEM	Cause and effect matrix	F	approval	15	10	IFR	PID, CNR, HAZ		
CMP	Commissioning procedure	B	approval	20	10	IFR	CAL, CNR, PID, PLN, SPC, STU		
CNR	Control narrative / functional description	F	review	15	10	IFR	PID, CNR		
CVL	Civil drawing (grading, roads, drainage, marine)	L	approval	20	10	IFR	PLT, STU, SDS	C	0.2
DBR	Design basis report	D	approval	20	10	IFR	-		
DSH	Datasheet	E	approval	10	5	IFR	PID, SPC	P	
EFD	Equipment foundation drawings (vendor-certified loads)	L	approval	15	10	IFR	CAL, FND	C	0.5
ELY	Electrical layout (earthing, lighting, trays, equipment)	L	review	20	10	IFR	ARC, CVL, ELY, FND, LAY, PPL, SLD, STU	C	0.2
FND	Foundation / underground drawing	L	approval	20	10	IFR	LAY, CAL	C	0.2
GAD	General arrangement drawing	L	review	20	10	IFR	ARC, DBR, LAY	C	0.3
HAZ	Safety study (HAZID, HAZOP, SIL, hazardous areas, fire, QRA)	S	approval	20	10	IFR	PID, PLT, HAZ		
HMB	Heat and mass balance	F	approval	15	10	IFR	DBR		
HMI	Operator display (HMI graphics) drawings	F	review	15	5	IFR	PID, CNR		
HUP	Instrument hook-up drawings	M	information	10	5	IFR	PID, SPC	C	0.5
IOL	I/O list	P	review	15	10	IFR	PID, CNR, LST, IOL		
ISO	Isometric drawing package	L	information	20	5	IFR	PPL	C	0.4
ITP	Inspection and test plan	Q	approval	10	5	IFR	SPC		
LAY	Equipment layout / arrangement	L	approval	20	10	IFR	ARC, HAZ, PLT	C	
LOG	Logic diagram	F	approval	20	10	IFR	CNR, IOL, RPT		
LOP	Loop / connection diagram	M	information	15	5	IFR	IOL	C	0.5
LST	List / index (equipment, lines, valves, instruments, loads, tie-ins)	P	review	15	10	IFR	DBR, HAZ, LST, PFD, PID, PRC, RPT, SLD, STU		
M3D	3D model review report (30 / 60 / 90 %)	L	review	15	5	IFR	PLT, LAY, PPL, ISO		
MDB	Manufacturing record book / conformity certificates / turnover dossier	Q	information	10	5	IFR	CMP, ITP		
MDL	Master document list	A	information	10	5	IFR	-		
MOS	Method statement / construction execution plan	B	review	10	5	IFR	CVL, PLN		
MRQ	Material requisition	C	information	5	3	IFR	SPC	P	
OMM	Operation and maintenance manual	D	review	30	10	IFR	CMP, PID		
PFD	Process flow diagram	F	approval	15	10	IFR	HMB		

Type	Title	IEC 61355	Review class	Prep	Upd.	Inputs at	Typical inputs	C/P	Share
PID	Piping and instrumentation diagram	F	approval	25	15	IFR	SDS		
PLN	Plan (execution, quality, HSE, commissioning, training)	B	approval	20	10	IFR	DBR, HAZ, LST, PFD, PLN, PRC, RPT, SPC, STU		
PLT	Plot plan	L	approval	20	10	IFR	DBR		
PPL	Piping general arrangement / plans	L	review	25	10	IFR	PID, LAY	C	0.2
PRC	Procedure / project standard	B	approval	15	10	IFR	-		
RBR	Reinforcement drawings	L	review	15	5	IFR	FND	C	0.3
RPT	Technical report	D	review	15	5	IFR	-		
SCH	Schedule	B	review	15	5	IFR	MDL		
SDS	System design description	D	review	15	10	IFR	DBR		
SLD	Single line diagram	F	approval	15	10	IFR	DBR, LST		
SPC	Technical specification	E	approval	20	10	IFR	-	P	
STL	Steel structure drawing	L	approval	20	10	IFR	GAD, CAL	C	0.3
STR	Pipe stress analysis report	E	review	20	5	IFR	PPL		
STU	Engineering study	D	review	25	10	IFR	-		
SUP	Pipe support drawings	L	information	15	5	IFC	ISO, STR	C	0.5
TRP	Test report / test record / certificate	Q	review	5	5	IFR	CMP, SPC, TST		
TST	Test procedure (FAT, SAT, site and performance tests)	Q	approval	15	10	IFR	DBR, ITP, RPT, SPC		
WIR	Schematic and wiring diagrams	F	information	15	5	IFC	SLD, LST	C	0.5

5. Number allocation

- Numbers are allocated only in the database: `python -m engine kks --next <ORG> <KKS> <DISC> <TYPE>` gives the next free number; the document is registered with `python -m engine db add document ... --reason`.
- Every document record carries `type_code`, discipline, originator and system consistent with its number; the `engineering_plan` engine checks all numbers at every run.
- A cancelled document keeps its number (status cancelled, purpose VOID); numbers are never reused.

6. Revisions and issue purposes

Purpose	Meaning	Revision
IFR	Issued for review / first issue	A, B, C ...
IFA	Issued for approval (approval class, after the Owner's comments are incorporated)	A, B, C ...
IFD	Issued for design (released internally for dependent design, e.g. vendor data frozen)	A, B, C ...
IFP	Issued for purchase (requisitions, specifications, datasheets)	A, B, C ...
IFC	Issued for construction / final	0, 1, 2 ...
IFI	Issued for information (information class, supplier documents received)	A, B, C ... or supplier rev.
AB	As built	next number
VOID	Cancelled; the number is never reused	-

Each issue is a `document_revision` record (revision, purpose, date, prepared / checked / approved by, file); packages are transmitted with `python -m engine deliver` (frozen copy and transmittal).

7. Review and approval

Class	Cycle
approval	Owner approval: IFR -> Owner review -> update -> IFA -> approval code 1 -> IFC
review	Owner review: IFR -> Owner review -> update -> IFC
information	sent to the Owner for information: IFR -> update -> IFC (no review period)
internal	consortium internal, not transmitted for review

Owner review period: 10 working days (14 calendar days, ER-17.01), 15 working days (21 calendar days) for Basic Design Package documents; after approval code 1 the IFC follows within 5 working days. Review codes:

Code	Meaning
1	No comments - proceed (approval: approved)
2	Proceed with the comments incorporated; re-issue
3	Do not proceed; revise and re-submit
4	For information / reference only; no review

Comments are recorded as `review_comment` records against the revision (comment -> response -> closed in the revision that incorporates it). The review class of a document type may be overridden per document (review field of the MDL line).

8. Supplier documents

- Imaginary Electric and subsuppliers submit per their VDRL, which is part of the MDL (originator IEC / subsupplier code) with committed first-issue and final dates.
- Supplier documents are numbered by Istanbul EPC before the purchase order, carry the project number as the primary number and are reviewed by Istanbul EPC before transmittal to the Owner where the class requires it.
- Received files are registered as source records (sources/) and linked to the MDL line (source field) with an IFR / IFI revision.

9. Master document list

The MDL is generated by the engineering_plan engine from the document records (2238 documents: C01 7, C02 7, C03 7, E01 14, E02 14, E03 11, E04 11, E05 12, E06 14, E07 12, E08 3, E09 3, E10 8, E11 8, E12 7, EPC 1488, H01 15, I01 12, I02 13, I03 14, I04 3, I05 3, I06 12, I07 12, I08 13, I09 12, I10 3, IEC 120, M01 15, M02 15, M03 15, M04 15, M05 16, M06 14, M07 15, M08 16, M09 16, M10 14, M11 14, M12 16, M13 15, M14 16, M15 15, M16 15, M17 7, M18 15, M19 15, M20 15, M21 13, M22 7, M23 7, M24 7, P01 3, P02 3, P03 3, P04 3, P05 3, P06 3, P07 3, S01 23, S02 8; 25 in the Basic Design Package). It is updated with every change of the records and issued monthly with the progress report (TQ-035).

Discipline	Documents	Hours	Sheets
civil	141	29,888	796
commissioning	242	7,910	265
electrical	184	16,506	3107
general	181	6,112	181
hse	38	3,924	59
hvac	22	4,850	156
i and c	425	28,480	9148
mechanical	656	13,418	2253
pipng	152	74,788	4075
process	153	18,007	402
project control	2	160	2
structural	42	8,282	951

10. Release control and process gates

A document is released in the order of its input network and only at the maturity its inputs allow; the processes that use the documents (procurement, Owner acceptance, construction, commissioning / acceptance, operations) start only through gates that name the documents and statuses they need. The rules below are applied by the engine: `python -m engine doc status <document>` shows the next purpose and revision and what blocks it; `python -m engine doc issue <document> --purpose <purpose>` refuses a blocked release (an --override is recorded on the revision); `python -m engine gate` lists the gates. The release_control engine issues the release plan, the relations, the impacts and the gate status (Excel).

Status	Reached when
IFR	first issue (also IFI / IFD / IFP)
IFA	issued for approval (approval class)
ACCEPTED	review code 1 or 2 on the latest issue; information and internal EPC documents when issued; supplier documents when accepted by Istanbul EPC
IFC	issued for construction / final (certified for supplier documents)
AB	as built

Release	Condition
IFR / IFA	every input at least at the maturity the document type requires (Inputs at, section 4); IFA after a reviewed IFR
IFC	every EPC input IFC, every supplier input ACCEPTED; the document ACCEPTED (approval / review class, supplier documents); no code 3
AB	the document is IFC
Impact	each revision records the input revisions it was based on; an input revised later makes the document CHECK REQUIRED, which blocks the release of the documents using it until it is re-issued or confirmed

Gate requirement: '[any:]selector[[selector][?]] STATUS' - selector type:<TYPE>, rule:<MDL rule>, id:<document>, bdp, construction, ewp-first, ewp-all; @mr = documents of the packages of the gate scope, @plant = plant-general documents, @epc / @supplier = originator; '|' unites several selectors, '?' = optional (none selected is not a finding), 'any:' = one selected document is enough. The plan uses the same gates: the PO is planned when the prerequisites of the PO gate are met, and the documents a milestone, key-date or CWP gate requires are prioritised to its date (ALP-EPC-00000-GE-PLN-0001).

Gate	Process	Step	Requires	Need	No.	Late	Float
G-PRC-ENQ	Procurement	Enquiry issued to bidders	type:SPC@epc IFR; type:MRQ@epc IFR	enquiry	59	0	5

Gate	Process	Step	Requires	Need	No.	Late	Float
G-PRC-FAB	Procurement	Manufacturing release (certified supplier data accepted)	rule:DL-V-GAD rule:DL-V-DSH? ACCEPTED; rule:DL-V-PID? ACCEPTED; rule:DL-V-CAL? ACCEPTED; rule:DL-V-FDL? ACCEPTED; rule:DL-V-ITP? ACCEPTED	fabrication	43	1	-4
G-PRC-FAT	Procurement	Factory acceptance test (procedure approved)	rule:DL-V-FAT ACCEPTED	shipment -20 wd	43	0	30
G-PRC-PO	Procurement	Purchase order award	type:SPC@epc IFC; type:MRQ@epc IFC; rule:DL-MR-TBE IFR; type:ITP@epc? IFR	po	59	0	0
G-PRC-SHP	Procurement	Release for shipment	rule:DL-V-FTR? ACCEPTED; rule:DL-V-MDB IFR	shipment	55	0	10
G-OWN-BDP	Owner acceptance	Basic Design Package approved (payment milestone MS-02)	bdp ACCEPTED	milestone:MS-02	1	1	-121
G-OWN-TO	Owner acceptance	Taking-Over: test reports accepted, O&M, spares, special tools, punch list	type:TRP@plant ACCEPTED; id:ALP-EPC-00000-CM-OMM-0001 IFC; id:ALP-EPC-00000-CM-LST-0002 IFC; type:LST@plant IFR	activity:KD-040	1	0	0
G-OWN-TOC	Owner acceptance	Tests on Completion procedures approved (90 days before the tests)	id:ALP-EPC-00000-CM-CMP-0001 ACCEPTED; id:ALP-EPC-00000-CM-CMP-0002 ACCEPTED; id:ALP-EPC-00000-CM-CMP-0003 ACCEPTED; id:ALP-EPC-00000-CM-CMP-0004 ACCEPTED; id:ALP-EPC-00000-CM-CMP-0005 ACCEPTED; id:ALP-EPC-00000-CM-CMP-0006 ACCEPTED	activity:KD-030 -65 wd	1	0	199
G-CON-CWP	Construction	CWP release: documents of the first IWP's IFC, installation manuals issued	ewp-first? IFC; any:ewp-all IFC; rule:DL-V-ERC@mr? IFR	cwp_start	59	0	4
G-CON-IWP	Construction	Progressive IWP documents IFC (isometrics, supports, schematics, loops ...)	ewp-all IFC	cwp_mid	59	0	12
G-COM-MC-EL	Commissioning	Mechanical completion of the system (before KD-010)	construction? IFC; rule:DL-CM-TOD IFR	activity:KD-010 -60 wd	18	0	50
G-COM-MC-FF	Commissioning	Mechanical completion of the system (before KD-020)	construction? IFC; rule:DL-CM-TOD IFR	activity:KD-020 -60 wd	53	0	30
G-COM-MC-WS	Commissioning	Mechanical completion of the system (before KD-030)	construction? IFC; rule:DL-CM-TOD IFR	activity:KD-030 -60 wd	20	0	50
G-COM-SC-EL	Commissioning	Start of system commissioning (before KD-010)	rule:DL-CM-CMP rule:DL-V-CMP@mr? ACCEPTED; rule:DL-CM-OPR rule:DL-V-OMM@mr? IFR; rule:DL-CM-TOD IFR	activity:KD-010 -40 wd	18	0	70
G-COM-SC-FF	Commissioning	Start of system commissioning (before KD-020)	rule:DL-CM-CMP rule:DL-V-CMP@mr? ACCEPTED; rule:DL-CM-OPR rule:DL-V-OMM@mr? IFR; rule:DL-CM-TOD IFR	activity:KD-020 -40 wd	53	0	50
G-COM-SC-WS	Commissioning	Start of system commissioning (before KD-030)	rule:DL-CM-CMP rule:DL-V-CMP@mr? ACCEPTED; rule:DL-CM-OPR rule:DL-V-OMM@mr? IFR; rule:DL-CM-TOD IFR	activity:KD-030 -40 wd	20	0	70
G-OPS-AB	Operations	As-built documentation, 3D model and engineering database (3 months after Taking-Over)	type:PID AB; type:SLD AB; type:ISO AB; rule:DL-PI-LAY AB; type:M3D@plant AB	activity:KD-040 +65 wd	1	0	-
G-OPS-HO-EL	Operations	Handover of the system to operations	rule:DL-CM-OPR rule:DL-V-OMM@mr? IFC; rule:DL-CM-TOD IFC	activity:KD-040	18	0	291
G-OPS-HO-FF	Operations	Handover of the system to operations	rule:DL-CM-OPR rule:DL-V-OMM@mr? IFC; rule:DL-CM-TOD IFC	activity:KD-040	53	0	192
G-OPS-HO-WS	Operations	Handover of the system to operations	rule:DL-CM-OPR rule:DL-V-OMM@mr? IFC; rule:DL-CM-TOD IFC	activity:KD-040	20	0	212

654 gate instances, 0 open now, 2 planned after their need date, 0 with a mandatory requirement selecting no document.

Gate	Scope	Need	Planned	Float wd	Driving document
G-OWN-BDP	00000	2027-02-02	2027-07-21	-121	ALP-EPC-20LCB-PR-PID-0001
G-PRC-FAB	MR-EL-011	2026-12-08	2026-12-14	-4	ALP-E11-00000-ME-ITP-0001

A gate planned late is resolved by resequencing (priority, staffing), expediting the supplier or a technical query; it stays a WARN of the release_control engine until the plan meets the need date.

- TQ-038 (open, raised 2026-09-28): Payment milestone MS-02 (Basic Design Package approved, month 3) cannot be met

Check results (engine)

1 OK, 0 INFO, 0 WARN (checks: numbering). WARN must be resolved before the next issue; INFO is noted.

Item	Check	Result	Detail
-	numbering	OK	2238 document numbers checked, 0 finding(s)